

Cybersecurity Temporal Behavioral Analysis: Advanced Threat Detection through Dynamic Time Warping and Honeypot Log Analysis

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Abstract

This research demonstrates that temporal patterns are a critical, overlooked dimension of threat intelligence. I introduce a novel analytical model that uses Dynamic Time Warping (DTW) and behavioral profiling to extract strategic intelligence from attack sequencing and pacing. Applying this to 12,002 honeypot sessions, we uncovered a landscape exclusively composed of hyper-efficient, fast-pattern attacks (avg. 14.4s).

The results shows two distinct types of attackers: a small group of persistent, repeat offenders and a much larger group of one-time scanners. I also discovered that attacks are over 5 times more frequent at night.

This work establishes that temporal behavioral analytics are transformative, moving cybersecurity from reactive detection to anticipating threats based on how and when they behave.

Discussion

The research findings have significant implications for cybersecurity strategy and operations. The complete dominance of fast-pattern attacks indicates an epidemic of automation and script-based exploitation, suggesting modern attackers prioritize speed and efficiency over traditional stealth approaches.

The clear bimodal threat distribution reveals two distinct attacker economies operating in parallel: professional operations conducting sustained campaigns and amateur actors performing basic reconnaissance. The strong nocturnal attack preference demonstrates sophisticated timing strategies designed to exploit reduced security monitoring during nighttime hours, particularly relevant for ransomware and data exfiltration operations.

The consistent command intensity across all threat levels further confirms the prevalence of automated toolkits, with human operators managing campaigns rather than executing individual attacks manually.

Results

The analysis reveals several critical insights into modern cyber attack patterns. The threat landscape demonstrates complete dominance of fast-pattern attacks, with all 12,002 sessions classified as rapid assaults averaging just 14.4 seconds duration per session.

These attacks maintain a consistent intensity of 1.00 command per second, achieving high efficiency with an average of 16.5 commands per session. The threat distribution shows a clear bimodal pattern, with 34.8% of attackers (223 IPs) classified as high-threat persistent actors averaging 52 sessions per IP, while 65.2% (417 IPs) represent medium-threat opportunistic scanners averaging only 1 session per IP.

Temporal analysis reveals strong nocturnal preferences, with peak activity occurring at 03:00 (13,180 events) and the quietest period at 12:00 (2,519 events), representing a 5.2-fold difference in attack volume.

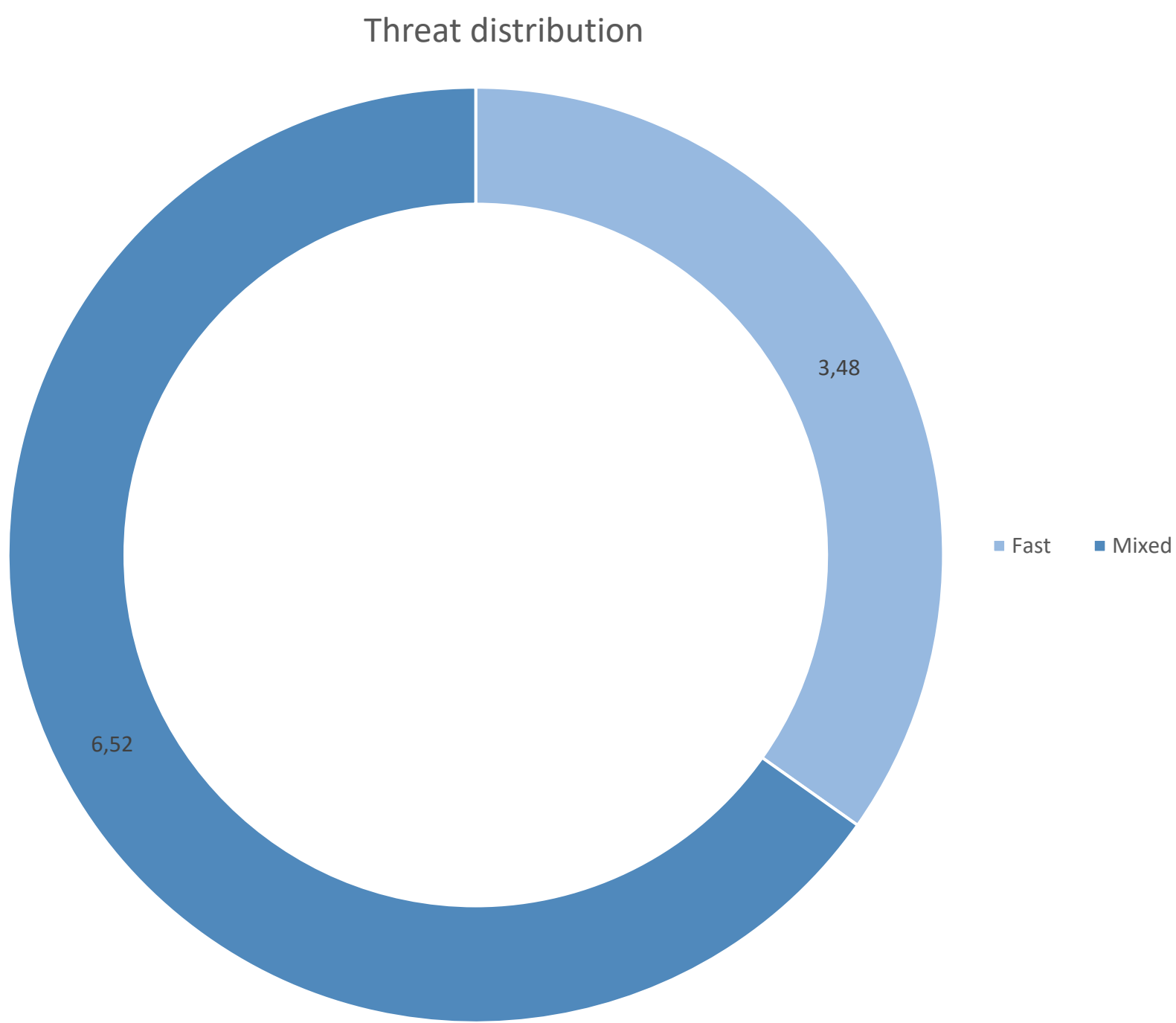


Figure 1. Threat distribution by intensity

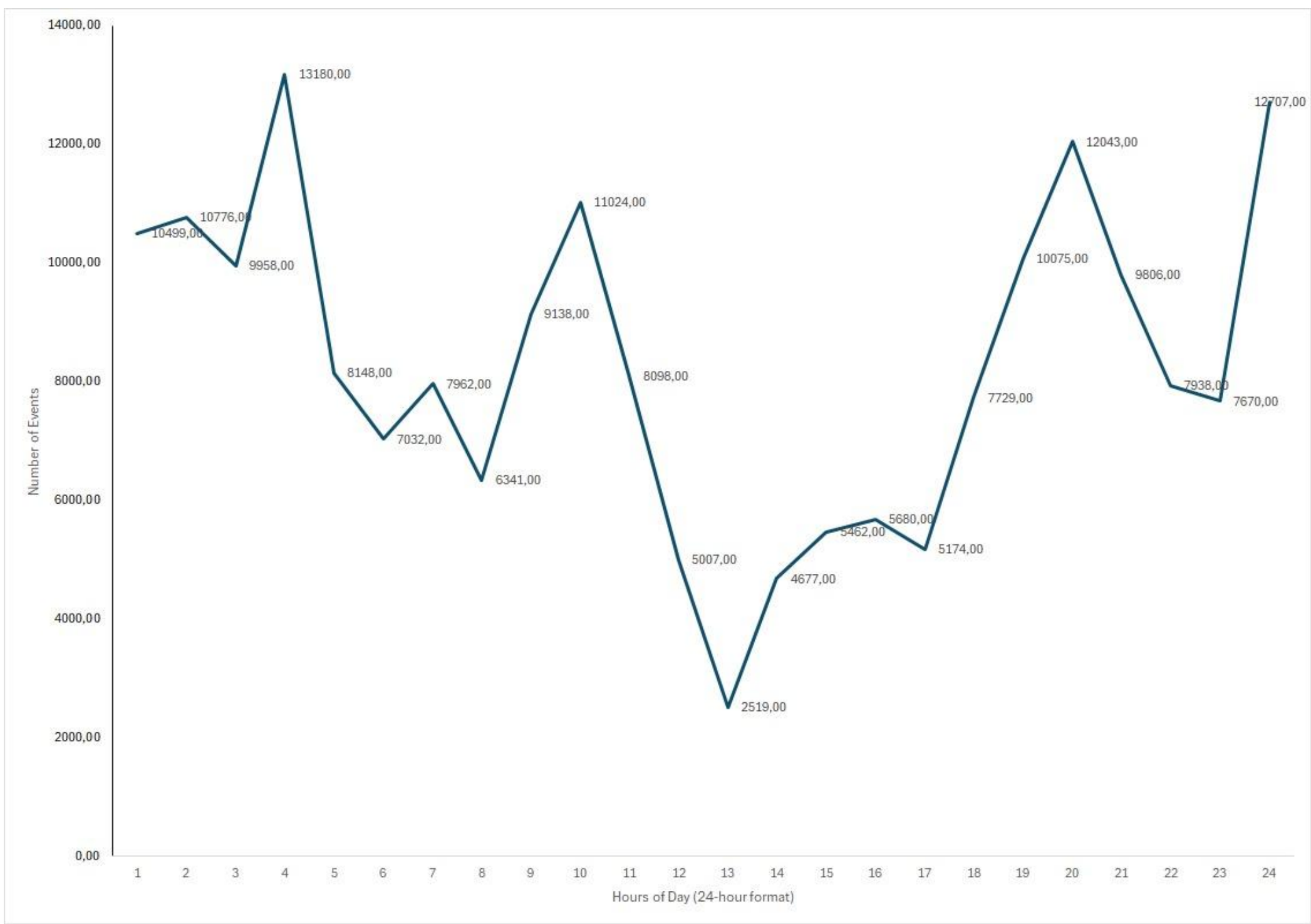


Figure 2. Events by hour of day.

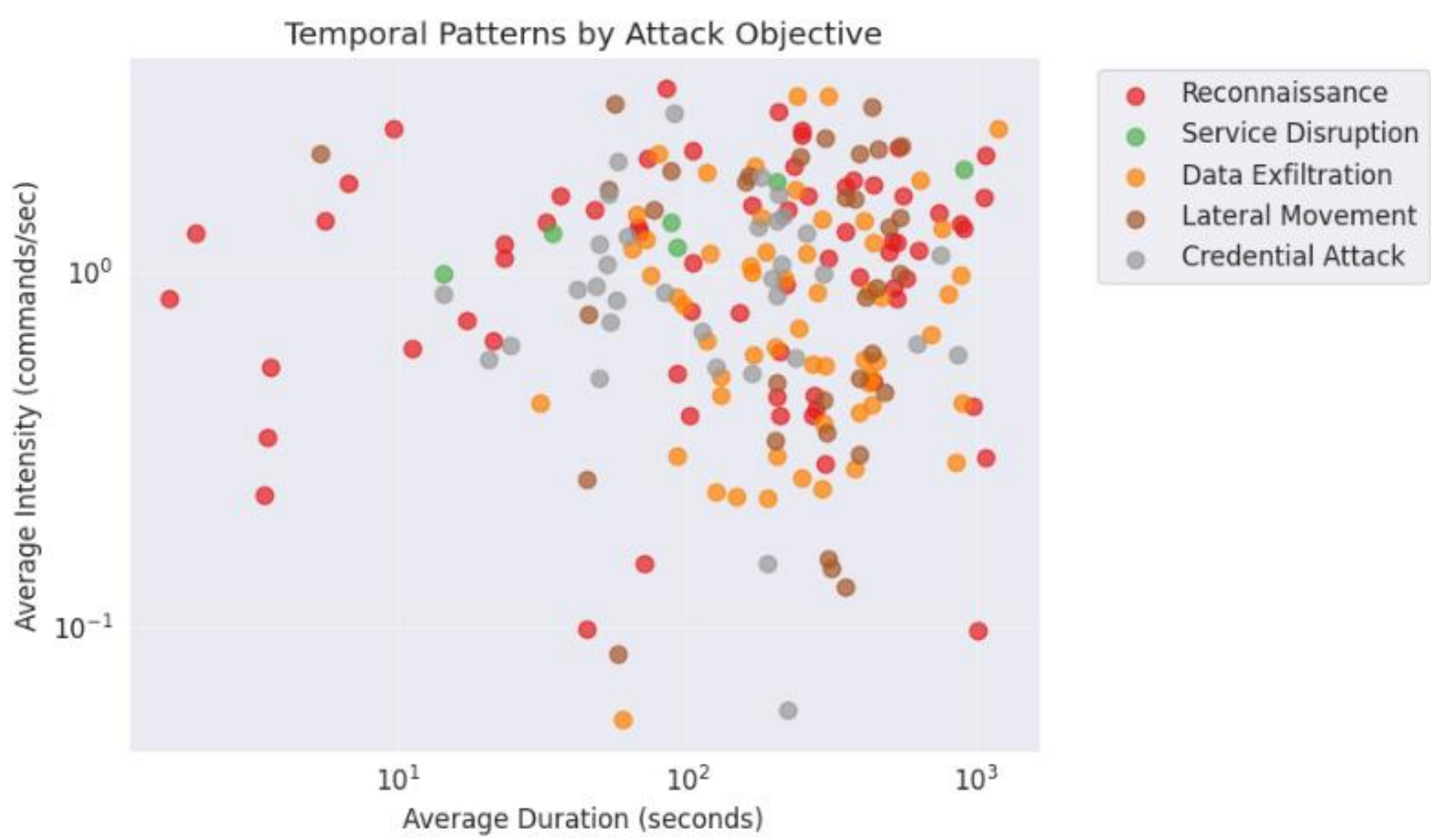


Figure 3. Temporal pattern by Attack objective

Introduction

The current landscape of cybersecurity analytics suffers from a critical limitation: most systems focus exclusively on what attacks occur, while completely neglecting the crucial temporal dimensions of how attacks progress over time.

This analytical gap leaves organizations vulnerable to sophisticated multi-stage attacks that strategically leverage timing as an evasion tactic. This research directly addresses this fundamental limitation by developing a comprehensive temporal behavioral analysis model that extracts strategic intelligence from attack sequencing, pacing, and consistency patterns.

The model moves beyond conventional approaches to provide deep insights into attacker methodologies and motivations through systematic analysis of temporal attack signatures.

Methodology

The research methodology employs a multi-stage analytical pipeline that begins with raw honeypot data collection and progresses through sophisticated behavioral analysis.



The DTW temporal analysis treats attacker sessions as time-series sequences, measuring behavioral similarity through optimal alignment between session patterns.

Behavioral profiling layer introduces multi-dimensional attack classification through speed pattern analysis and threat categorization. The system employs statistical quartile analysis to classify session tempo into Fast, Slow, and Mixed categories based on commands-per-second metrics.

The advanced analytics layer integrates MITRE ATT&CK framework mapping with Dynamic Time Warping (DTW) for sophisticated threat behavior analysis.

Conclusions

This research establishes that temporal behavioral analytics transform conventional cybersecurity from reactive detection to predictive threat intelligence. By analyzing attack patterns across 12,002 sessions, demonstrated how timing signatures reveal attacker objectives and automation levels.

The dominance of fast-pattern attacks and nocturnal activity peaks provides actionable intelligence for resource allocation and proactive defense. This methodology enables organizations to anticipate threats based on behavioral patterns, fundamentally advancing how we understand and counter evolving cyber threats.

References

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